Online Appendix

The Price of Protecting Small Firms: Evidence from Public Procurement

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This Online Appendix provides supplementary results for "The Price of Protecting Small Firms: Evidence from Public Procurement." All table and figure numbers are prefixed with OA to distinguish them from the main text.

1 Raw Monthly Trends

Figures OA.1 through OA.4 display the unconditional monthly means of each outcome variable, separately for Group 65 and all other groups. The raw trends are broadly consistent with the event study estimates, showing a narrowing of the gap between the two groups after the policy change.

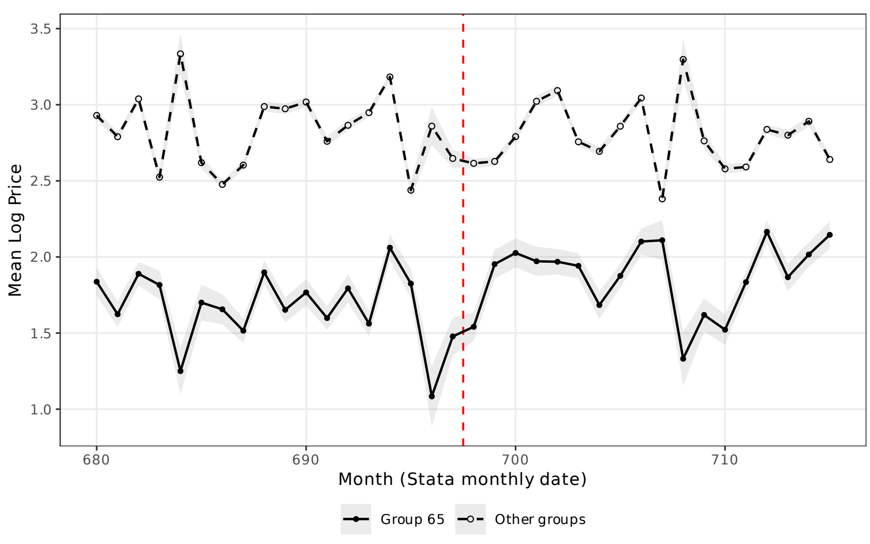

Figure OA.1: Raw Trends: Mean Log Price by Month (Group 65 vs. Other Groups)

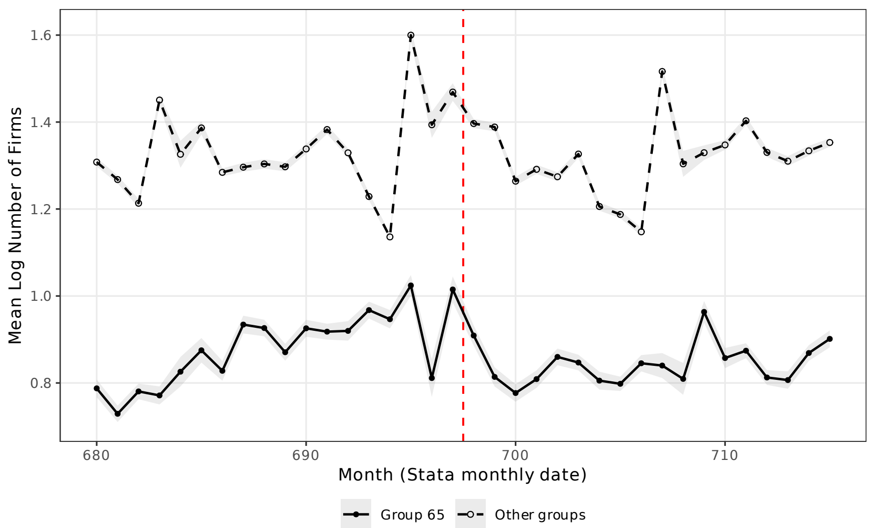

Figure OA.2: Raw Trends: Mean Log Number of Firms by Month (Group 65 vs. Other Groups)

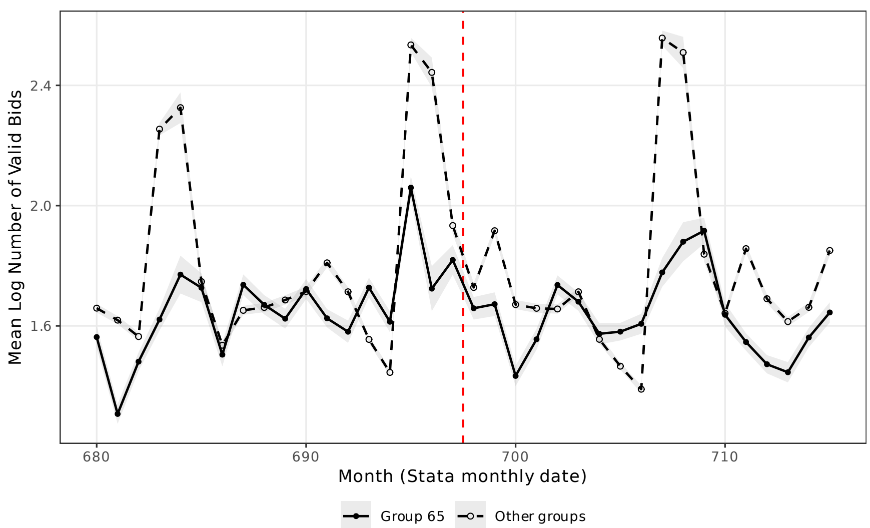

Figure OA.3: Raw Trends: Mean Log Number of Valid Bids by Month (Group 65 vs. Other Groups)

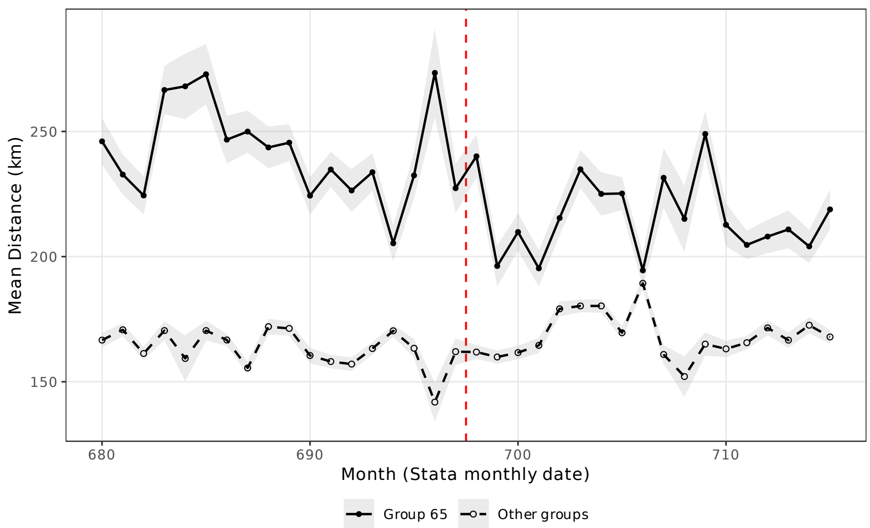

Figure OA.4: Raw Trends: Mean Distance (km) by Month (Group 65 vs. Other Groups)

2 Randomization Inference

Figure OA.5 presents the results of a randomization inference exercise. The two-sided permutation p-value is 0.317, reflecting the inherently low power of permutation tests when treatment is assigned to a single unit out of 76 product groups.

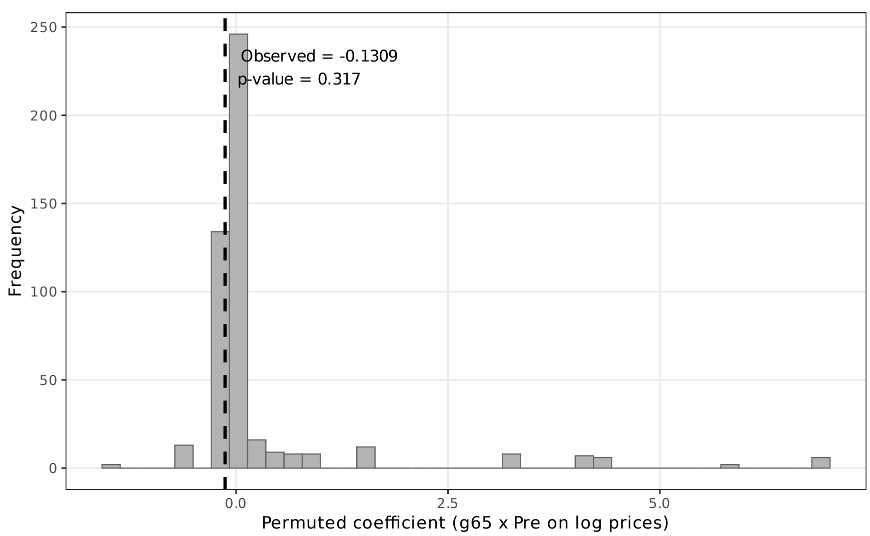

Figure OA.5: Randomization Inference: Distribution of Permuted Coefficients (Log Prices, 18-month window)

3 SME Participation Trends

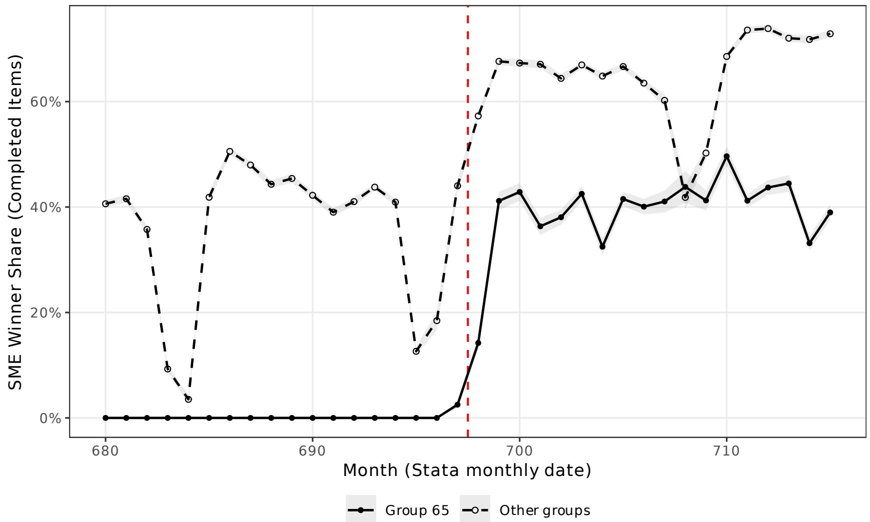

Figure OA.6: SME Winner Share (Completed Items) by Month

4 Extension Analyses

Table OA.1: Real Prices (IPCA-deflated, log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.1077***	-0.1210***	-0.0994***	-0.0997***	-0.0762***	-0.0795***
	(0.0121)	(0.0117)	(0.0108)	(0.0104)	(0.0096)	(0.0094)
Sealed bids	-0.1464***	-0.2225***	-0.1422***	-0.2029***	-0.1490***	-0.2059***
	(0.0088)	(0.0131)	(0.0079)	(0.0123)	(0.0079)	(0.0116)
lquantity	-0.2603***	-0.2978***	-0.2622***	-0.2966***	-0.2609***	-0.2944***
	(0.0082)	(0.0089)	(0.0075)	(0.0080)	(0.0073)	(0.0077)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1911	0.2128	0.1936	0.2115	0.1941	0.2112
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table OA.2: Extensive Margin: Tender Completion Rate

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	0.1165***	0.1179***	0.1244***	0.1265***	0.1072***	0.1133***
	(0.0044)	(0.0044)	(0.0035)	(0.0036)	(0.0032)	(0.0032)
Sealed bids	0.0735***	0.0607***	0.0638***	0.0486***	0.0651***	0.0544***
	(0.0025)	(0.0033)	(0.0020)	(0.0025)	(0.0018)	(0.0023)
lquantity	0.0224***	0.0256***	0.0258***	0.0284***	0.0251***	0.0274***
	(0.0007)	(0.0007)	(0.0006)	(0.0006)	(0.0005)	(0.0005)
Observations	304,392	304,392	609,299	609,299	901,506	901,506
R-squared	0.0181	0.0177	0.0211	0.0209	0.0195	0.0194
Group FE	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: DV = 1 if item was successfully purchased, 0 otherwise. All items included (not filtered to completed). Group FE used instead of item FE. Standard errors clustered at the item level. *** p<0.01, ** p<0.05, * p<0.1.

Table OA.3: Price Efficiency: Final Price / Reference Price (%)

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.0651***	-0.0688***	-0.4399	-0.5009	-0.3033	-0.3462
	(0.0115)	(0.0109)	(0.3740)	(0.4354)	(0.2566)	(0.3010)
Sealed bids	-0.0620***	-0.0501***	0.0039	0.0605	-0.0218	0.0113
	(0.0027)	(0.0044)	(0.0608)	(0.1062)	(0.0371)	(0.0623)
lquantity	-0.0148***	-0.0155***	0.1277	0.1382	0.0755	0.0804
	(0.0018)	(0.0018)	(0.1426)	(0.1521)	(0.0909)	(0.0962)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.0038	0.0025	0.0001	0.0001	0.0001	0.0001
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Constant absorbed by fixed effects.

Table OA.4: Winner Composition: SME Winner (binary)

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.3388***	-0.3615***	-0.3633***	-0.3771***	-0.3858***	-0.4056***
	(0.0053)	(0.0053)	(0.0044)	(0.0044)	(0.0040)	(0.0039)
Sealed bids	0.4126***	0.3882***	0.4614***	0.4341***	0.4190***	0.3918***
	(0.0043)	(0.0053)	(0.0037)	(0.0039)	(0.0032)	(0.0032)
lquantity	0.0096***	0.0141***	0.0090***	0.0126***	0.0062***	0.0104***
	(0.0010)	(0.0008)	(0.0008)	(0.0006)	(0.0007)	(0.0006)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1410	0.1267	0.1746	0.1486	0.1583	0.1333
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table OA.5: Heterogeneous Effects by PBU Type (18-month window)

	Log prices		Log firms	
	Base	+PBU FE	Base	+PBU FE
$g65 \times Pre$	-0.0718***	-0.0697***	-0.0146	0.0567***
	(0.0231)	(0.0230)	(0.0113)	(0.0120)
Direct admin.	0.0425***	–	0.0421***	–
	(0.0084)		(0.0044)	
$g65 \times Pre \times$ Direct admin.	-0.0655***	-0.0685***	0.1175***	0.0475***
	(0.0236)	(0.0227)	(0.0117)	(0.0122)
Observations	649,714	649,714	773,263	773,263
R-squared	0.1936	0.2108	0.2065	0.1632
Item FE	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES

Notes: 18-month window. Direct admin. is 1 for PBUs under direct administration, 0 otherwise. Standard errors clustered at the item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

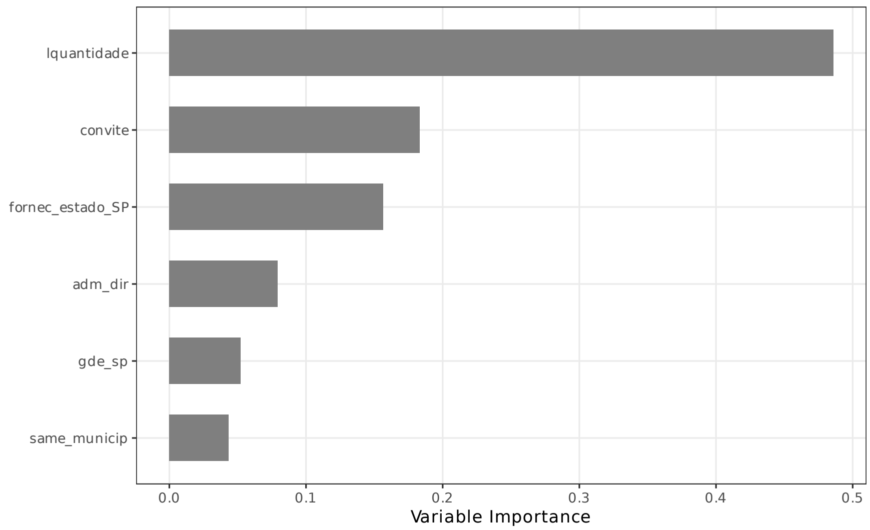

Figure OA.7: Causal Forest: Variable Importance for Treatment Effect Heterogeneity

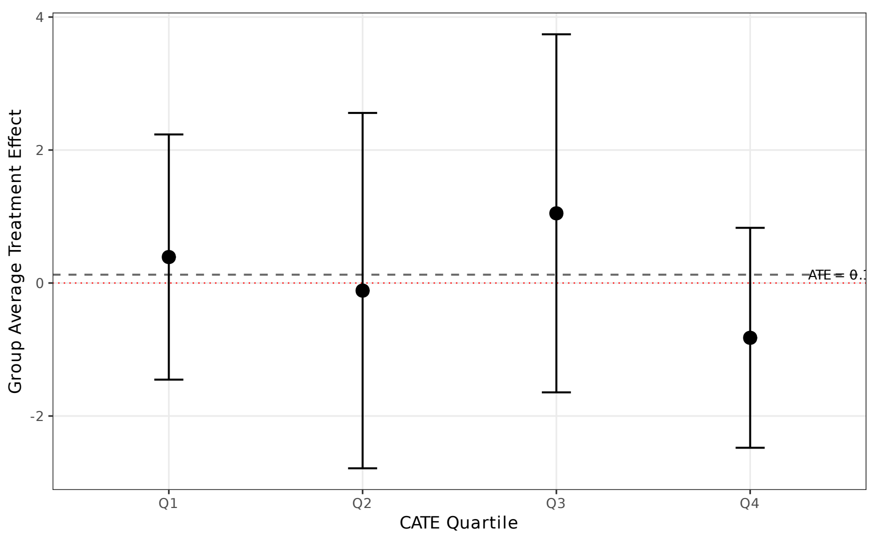

Figure OA.8: Causal Forest: Group Average Treatment Effects by CATE Quartile

Table OA.6: Causal Forest: Group Average Treatment Effects by CATE Quartile

	Q1 (lowest)	Q2	Q3	Q4 (highest)
GATE	0.3906	-0.1143	1.0482	-0.8240
	(0.9407)	(1.3634)	(1.3738)	(0.8438)
ATE (full sample)	0.1251 (0.5779)			

Notes: Causal forest estimated with 2,000 honest trees on FWL-residualized outcomes. CATE quartiles defined by predicted individual treatment effects. Top variable importances: lquantidade: 0.486, convite: 0.183, fornec_estado_SP: 0.156, adm_dir: 0.079. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

5 Heterogeneous Treatment Effects (Causal Forest)

6 Distributional Effects (Quantile DiD)

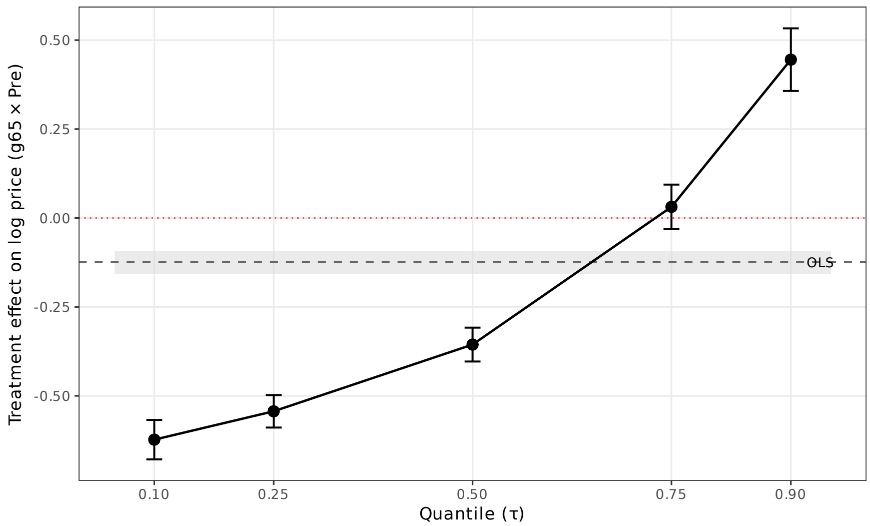

Figure OA.9: Quantile Difference-in-Differences: Treatment Effects Across the Price Distribution

Table OA.7: Quantile Difference-in-Differences: Treatment Effects Across the Price Distribution

	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
$g65 \times Pre$	-0.6231***	-0.5434***	-0.3558***	0.0313	0.4451***
	(0.0283)	(0.0233)	(0.0243)	(0.0319)	(0.0449)
OLS benchmark			-0.1241 (0.0165)		
Observations			100,000		
Group FE			YES		

Notes: Quantile regression estimated at each τ using the Canay (2011) two-step estimator with group FE (78 levels). 18-month window, completed items. OLS benchmark estimated on identical sample with group FE. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

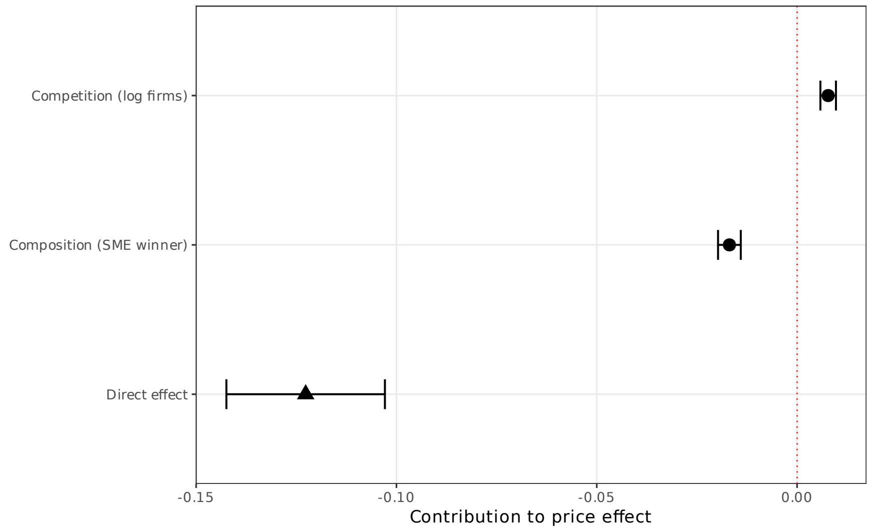

Figure OA.10: Gelbach (2016) Decomposition: Channel Contributions to the Price Effect

Table OA.8: Gelbach (2016) Decomposition: Channels of Price Effect

	Coefficient	SE	% of gap
Panel A: Overall effect			
Short regression ($g65 \times Pre$)	-0.1318***	(0.0096)	
Full regression ($g65 \times Pre$)	-0.1227***	(0.0101)	
Gap (short − full)	-0.0091		100.0%
Panel B: Channel decomposition ($\delta_k = \gamma_k \times \pi_k$)			
Competition (log firms)	0.0078***	(0.0010)	-85.0%
Composition (SME winner)	-0.0169***	(0.0014)	185.0%
Direct effect (unexplained)	-0.1227***	(0.0101)	
Observations		648,616	
Item FE		YES	

Notes: Gelbach (2016) decomposition of the price effect into channel contributions. Short regression includes only treatment and controls; full regression adds mediators. $\delta_k = \gamma_k \times \pi_k$ where γ_k is the mediator coefficient in the full regression and π_k is the treatment coefficient in the auxiliary regression of mediator k on treatment. 18-month window, completed items. SE via delta method. *** p<0.01, ** p<0.05, * p<0.1.

Table OA.9: Balance Test: Pre-Treatment Covariate Means by Group

	Group 65		Other Groups			
	Mean	SD	Mean	SD	Difference	p-value
Log quantity	5.08	2.89	3.10	2.27	1.985***	0.000
Sealed bid (share)	0.22	0.42	0.60	0.49	-0.377***	0.000
Completion rate	0.67	0.47	0.77	0.42	-0.101***	0.000
Number of firms	3.17	2.52	4.66	3.27	-1.482***	0.000
Number of valid bids	11.10	15.49	10.13	15.74	0.970***	0.000
Price (levels)	1,402.09	20,563.54	1,465.85	50,385.93	-63.755	0.622
Distance (km)	238.54	270.42	164.78	182.23	73.761***	0.000

Notes: Pre-treatment period means (September 2016 to February 2018) for Group 65 and all other groups. Difference = Group 65 − Others. p-values from Welch two-sample t-tests. Price and distance are conditional on item completion. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Mechanism Decomposition (Gelbach)

8 Pre-Treatment Balance Test

9 Excluding Last Post-Period Semester

Table OA.10: Robustness: Excluding Last Post-Period Semester (Mar–Aug 2019)

	Log prices		Log firms		Log bids		Distance	
	Base	PBU FE	Base	PBU FE	Base	PBU FE	Base	PBU FE
$g65 \times Pre$	-0.1300***	-0.1297***	0.1031***	0.1091***	0.0512***	0.0608***	9.6580***	4.7026**
	(0.0116)	(0.0113)	(0.0063)	(0.0064)	(0.0082)	(0.0083)	(2.3462)	(2.3492)
Observations	528,676	528,676	629,390	629,390	629,390	629,390	528,676	528,676
R^2	0.9206	0.9256	0.5570	0.5955	0.5639	0.5945	0.3359	0.4267

Notes: Re-estimation of the 18-month window specification excluding the final post-period semester (March–August 2019). The effective window is September 2016 to February 2019. Standard errors clustered at the item level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

10 Mechanism Evidence

11 Synthetic Control

References

Table OA.11: Dose-Response: Effects by Item Quantity

	Log prices		Log firms	
	Base	PBU FE	Base	PBU FE
Panel A: High quantity (above median)				
$g65 \times Pre$	-0.1230***	-0.1297***	0.1165***	0.1265***
	(0.0111)	(0.0108)	(0.0070)	(0.0071)
Observations	348,363	348,363	402,194	402,194
Panel B: Low quantity (below median)				
$g65 \times Pre$	-0.1336***	-0.1289***	0.0235***	0.0234***
	(0.0117)	(0.0111)	(0.0077)	(0.0077)
Observations	301,351	301,351	371,069	371,069

Notes: 18-month window. The sample is split at the median of log quantity (3.18). High-quantity items are more likely to attract large non-SME suppliers, so the SME restriction is more binding for these items. This split aligns with the causal forest finding that item quantity is the dominant source of treatment effect heterogeneity (variable importance = 0.486). SE clustered at item level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.12: Mechanism Decomposition: Entry Margin and Winner Composition

	(1) Log price	(2) Log firms	(3) SME winner	(4) Price + firms	(5) Price + SME	(6) Price + both
$g65 \times Pre$	-0.1309***	0.0926***	-0.3858***	-0.1395***	-0.1159***	-0.1227***
	(0.0096)	(0.0059)	(0.0040)	(0.0099)	(0.0098)	(0.0101)
Log firms				0.1087***		0.1099***
				(0.0106)		(0.0106)
SME winner					0.0390***	0.0438***
					(0.0037)	(0.0037)
Observations	649,714	773,263	649,714	648,616	649,714	648,616

Notes: 18-month window, item FE, SE clustered at item level. Columns (1)–(3): DiDiR for each outcome separately. Column (4): price regression adding log firms as control (mediation by competition). Column (5): price regression adding SME winner indicator (mediation by composition). Column (6): price regression with both mediators. The attenuation of the $g65 \times Pre$ coefficient from (1) to (4)–(6) measures the fraction of the price effect explained by each channel. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.13: Within-Group-65 Heterogeneity: Medications vs. Other Medical Supplies

	Log prices		Log firms	
	Base	PBU FE	Base	PBU FE
Panel A: Medications (class 6531)				
$g65_{med} \times Pre$	-0.1658***	-0.1744***	0.1516***	0.1605***
	(0.0174)	(0.0172)	(0.0100)	(0.0101)
Observations	573,080	573,080	678,547	678,547
Panel B: Other medical supplies (non-6531)				
$g65_{other} \times Pre$	-0.0992***	-0.0973***	0.0444***	0.0504***
	(0.0092)	(0.0086)	(0.0057)	(0.0057)
Observations	593,920	593,920	705,616	705,616

Notes: 18-month window. Panel A restricts group 65 to class 6531 (medications, subject to CMED pharmaceutical price regulation), using all non-group-65 items as the comparison. Panel B restricts group 65 to non-medication medical supplies (hospital equipment, dental supplies, etc.). If the price effect were driven by differential pharmaceutical inflation rather than competition, it should appear primarily in Panel A. SE clustered at item level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

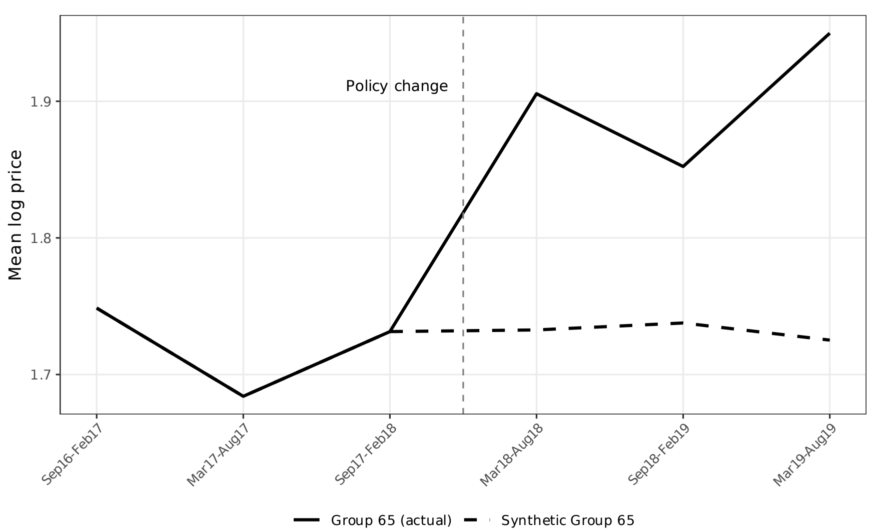

Figure OA.11: Synthetic Control: Group 65 (Actual) vs. Synthetic Group 65

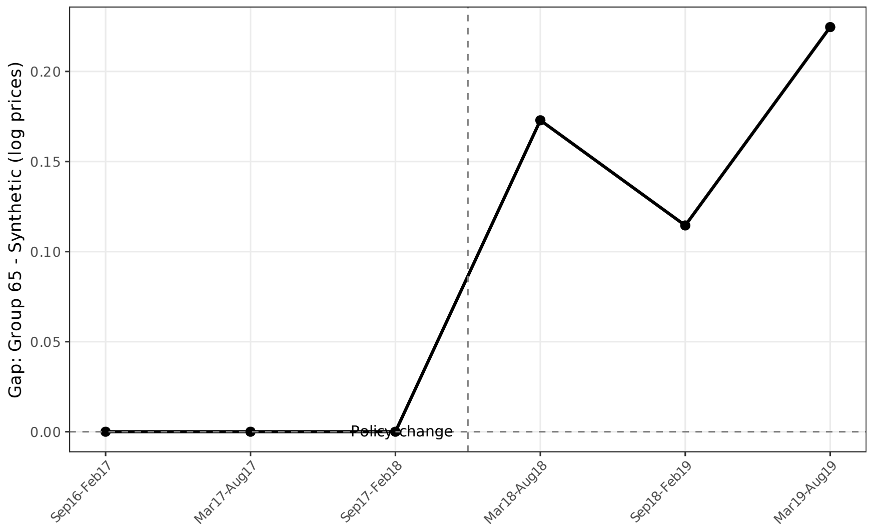

Figure OA.12: Synthetic Control: Gap Between Group 65 and Synthetic (Log Prices)

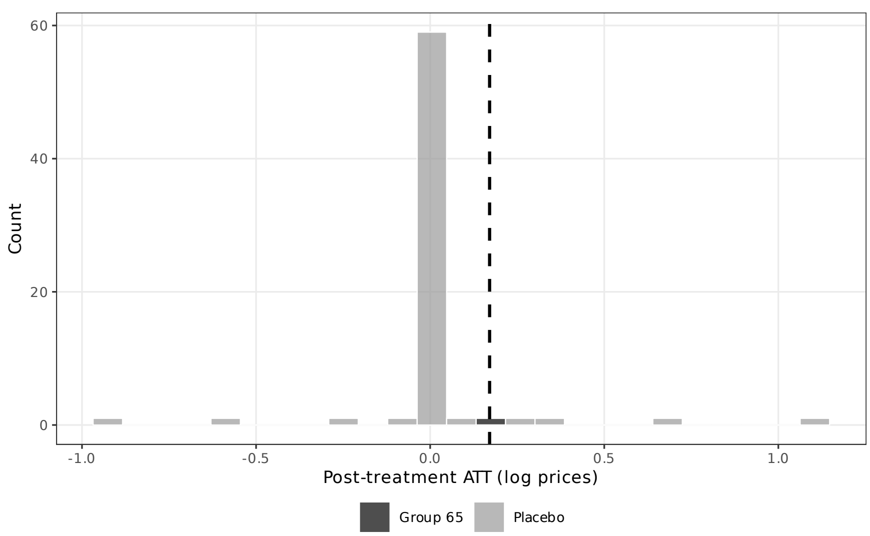

Figure OA.13: Synthetic Control: In-Space Placebo Distribution of Post-Treatment ATTs

Table OA.14: Synthetic Control: Summary Results

Panel A: Gaps	
Mean pre-treatment gap (semesters 1–3)	0.0000
Mean post-treatment gap (semesters 4–6)	0.1707
Post-treatment ATT (mean)	0.1707
Placebo rank p-value	0.103 (7/68)
L2 imbalance	0.0000
Panel B: Top donor weights (> 1%)	
Product group	Weight
Group 40	0.044
Group 75	0.099
Group 79	0.382
Group 89	0.436
Other groups	0.039
Balanced groups	69
Semesters	6

Notes: Augmented synthetic control with Ridge augmentation. Outcome: mean log price at the group $\times$ semester level. Treatment date: March 2018 (group 65 loses open-tender exemption). Pre-treatment gap measures how much lower group 65 prices were under open tenders relative to the synthetic counterfactual. Post-treatment gap near zero validates the parallel trends assumption. Inference via conformal method.